GE Medical Systems

Technical Publication

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Revision 2

CT DAS Cleaning and De-ionizing Procedure
SNT-1434

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REVISION HISTORY

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Revision	Date	Reason for change
0	05/16/00	Initial Release
1	03/26/01	Added warm-up spec to retest; Added kit # for MDAS
2	06/19/02	CQA 1024060 -- Update Kit part numbers

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CT DAS CLEANING AND DE-IONIZING PROCEDURE

OBJECTIVE OF THE PROCEDURE

The following procedure should be followed when DAS Converter boards and/or chassis needs cleaning or Converter boards are replaced for microphonics or noise resulting in image artifacts (rings, bands, streaks and smudges).

TOOLS NEEDED

1. Static Wrist Strap and cord
2. Lint free Towels. GE Part Number: 46-307237P1 or equivalent
3. Amax Contact, and Circuit Cleaner, Part #362-16, MSDS #:8365595

   GE Part Number: 2274104
4. DAS/DET Interface Kit: GE Part Number 2227089-2
5. Aero Duster Part # 2226685 (MS-222N)
6. Screw driver (flat) size # 6

WARNING

Rotating Gantry can cause severe injury or death, make sure Gantry is properly de-energized and locked while performing this service. Follow the Gantry Safety Instructions in Chapter 1 of the System Service Manual.

Mishandling can easily damage converter cards. Handle only on the sides and only with proper ESD protection. Do not touch the connector. Cards should always be placed in a Static Bag when not in the DAS

Store Amax Cleaner at Site with MSDS document. Do not store in the company car. Use spray as directed in this precedure. Safety glasses and Nitrile gloves must be worn when using the spray.

PROCEDURE

Take DC Noise baseline scan using DASTools Manual test (use default settings) to establish the DAS' current noise characteristics. Do not troubleshoot any noise failures until after the cleaning.

1. Use Scan Analysis and plot the means and standard deviations for all rows to find channels that violate the noise specification. Refer to System Service Manual for the acceptable noise values. The Channel Map tool can be used to determine which channel(s) is (are) noisy and the card location.
2. Lock out system power (See System Service Manual, Chapter 1)
3. Open Gantry and shut off 550V, axial drive, and DAS power.
4. All protective ESD materials should be in place, i.e. wrist straps and grounded mats for laying out converter cards.
5. Rotate and lock the gantry so that the DAS Chassis that you are working with is vertical so that run off from the Spray cleaner does not get into the elastomer housings. See Figure 1.

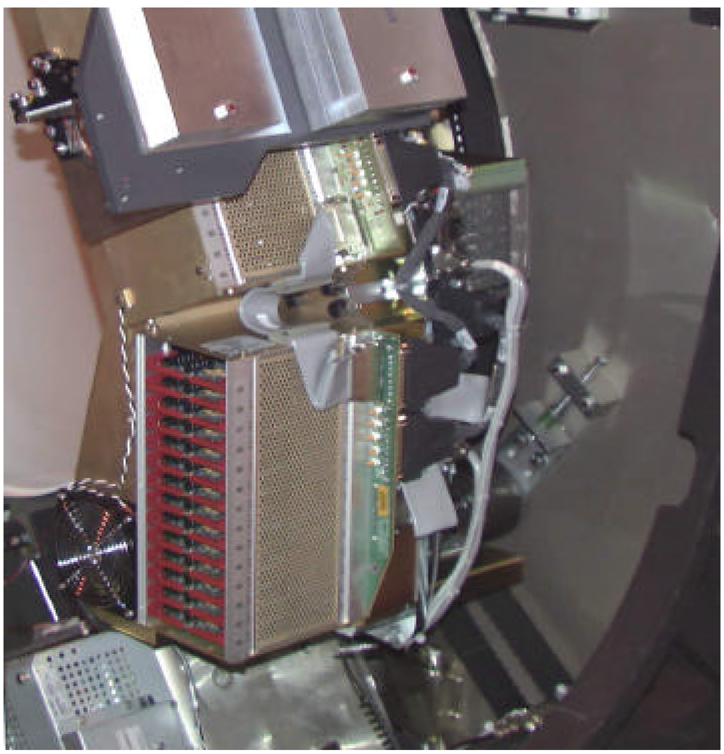

Figure 1 - Proper Positioning of DAS Chassis for Cleaning

6. Use the #6 screwdriver to remove the cover of the DAS chassis that has the suspected bad card(s).
7. Remove the suspected noisy Converter card(s) and place it (them) in static free bags. Also remove the neighboring cards to the “bad” card and place them in static free bags. Mark down the slot positions of all removed cards so they can be put back in the same slots after cleaning.

8. At this point, examine the DAS for dust. If there is dust in the DAS, perform the inspection and cleaning procedure as prescribed in the Job Card for DAS Cleaning which is in the Advanced Service Manual. If there are still noisy channels after completing the DAS Cleaning procedure, repeat this cleaning procedure starting from step 1. If there was not any dust inside the DAS, continue with this procedure.
9. Install the Aero Duster Spray System on the Aero Duster can and spray off the backplane connector from which the cards were removed. Use the Aero Duster Spray System every time the compressed air is used in this procedure.
10. Position "lint free" towels between the chassis and the flex housings. This will prevent any excess cleaner entering any parts not needing cleaning. See Figure 2.

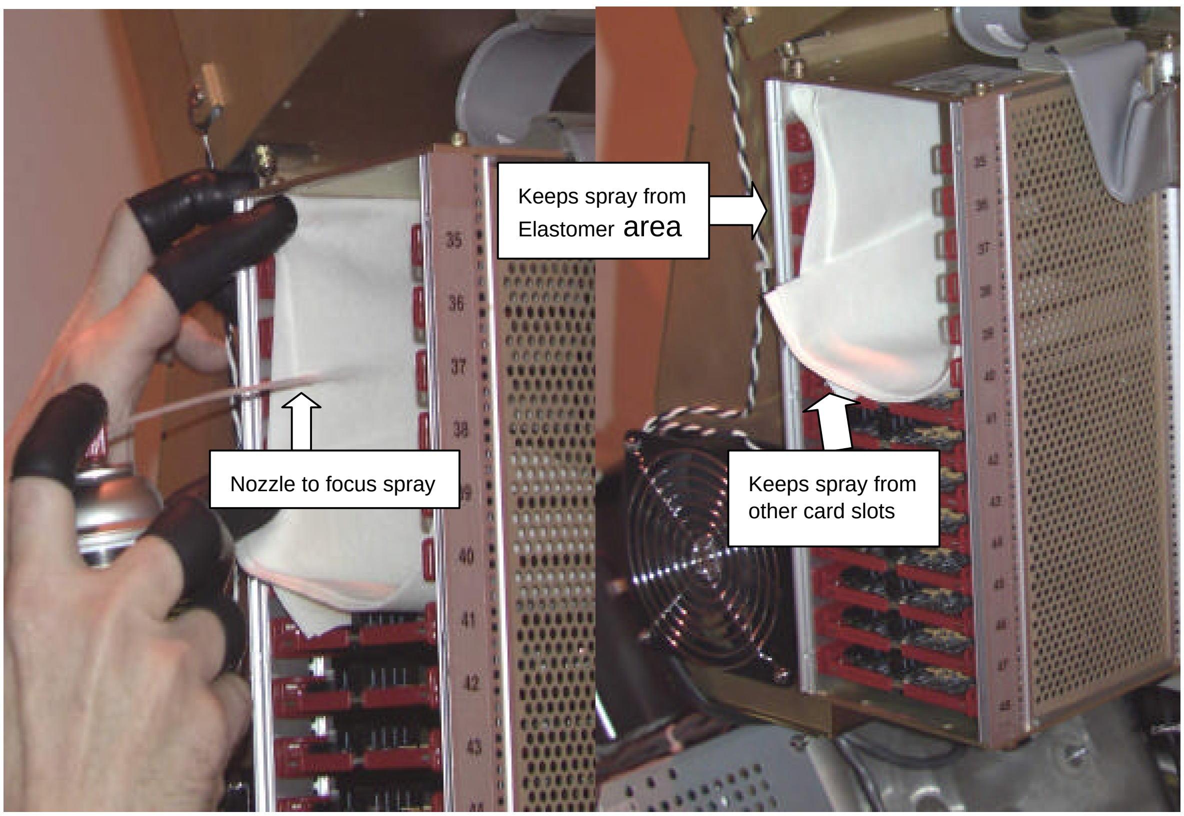

Figure 2 - Proper placement of "lint-free" towels

11. Use the Amax Contact Cleaner spray with the plastic tube to focus the spray to clean the backplane connectors of the suspected “bad” card location. Apply only enough to wet the entire backplane connector(s). Amax Contact Cleaner dries extremely fast. Avoid spraying directly on the detector or elastomers.
12. Let the backplane dry for 2 minutes. Do not spray air in the chassis to dry the cleaner.
13. Remove the Converter cards (see Figure 3) one at a time from their static bags and spray the connector (see Figure 4) with it facing down (see so any spray will drip off the card and not across the converter card. Spray the outside of the connector shroud on all sides and also spray inside the pin housing. (Give it a good soaking). Allow the card to air dry. Do not use the the compressed air to dry it.

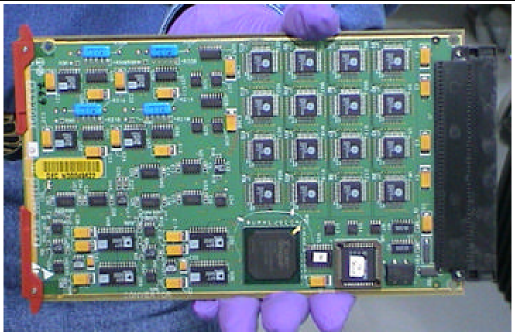

Figure 3 - Proper method for handling converter cards

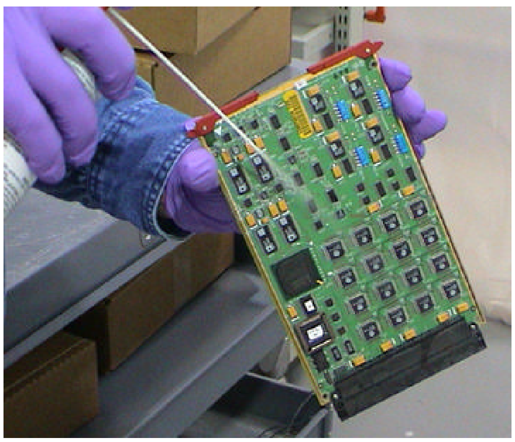

Figure 4 - Spraying "off" the connectors

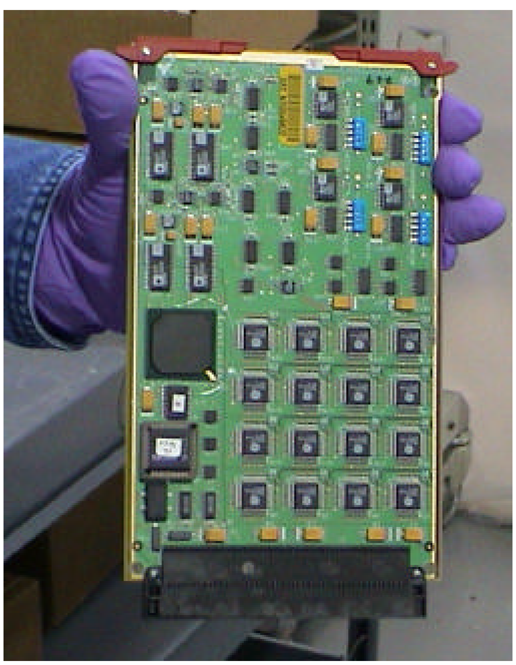

Figure 5 - Allow the cleaner to run-off the converter card completely.

14. Reinstall cards in the same slots from which they were taken.

    Turn the DAS ON and let it warm up. It takes two (2) minutes of warm-up time for every minute the DAS was turned OFF, up to one (1) hour.

    Verify all covers are installed, turn on DAS power and complete the following tests within DASTools.

    - 1 iteration of DC Means/Noise
    - 1 iteration of microphonics/pop

    All tests must pass. If not, troubleshoot and correct the failures. Refer to DAS Troubleshooting job cards for reference.

    Verify Image Quality by performing the 1x Image Series found in the System Service manual.

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